

Audit Trail — Case 11: Shoulder closure on divided highway

Field	Value
Plan status	1 change · 1 needs attention · 14 checked · 2 pending · reference
MUTCD typical application	TA-3
CDOT standard sheet	S-630-1
Case	Case 11: Shoulder closure on divided highway
Taper length	150 ft (L/3 (shoulder taper))
Buffer space	360 ft
Device spacing — taper	45 ft
Device spacing — tangent	90 ft
Crew steps	14
Routing	shoulder_no_reduction

Taper Length

Speed 45 mph $\geq$ 45 mph threshold -> using $L = W \times S$

$$L = 10 \times 45 = 450 \text{ ft}$$

$$L/3 = 450 / 3 = 150 \text{ ft}$$

Field	Value
Closure type	shoulder
Offset (shoulder width)	10 ft
Required taper (L/3 (shoulder taper))	150 ft

Source: MUTCD 11th Ed. Sec 6B.08, Table 6B-4 (taper length L). Shoulder closures use L/3 per Sec 6B.08 (Table 6B-3).

CDOT reference: CDOT S-630-1 Case 11 (right-shoulder closure on divided highway)

Buffer Space

MUTCD Table 6B-2: 360 ft (CDOT supplement silent at this speed)

Source: MUTCD 11th Ed. Sec 6B.06, Table 6B-2 (stopping sight distance)

Channelizing Device Spacing

45 mph -> 45 ft spacing (MUTCD Sec 6K.01: spacing equals speed in feet)

45 mph -> 90 ft spacing (MUTCD Sec 6K.01: spacing equals 2x speed in feet)

$L/3 = 150 \text{ ft} / 45 \text{ ft max spacing} \rightarrow 5 \text{ drums at } 37.5 \text{ ft intervals}$

$1000 \text{ ft} / 90 \text{ ft max spacing} \rightarrow 13 \text{ cones at } 83.3 \text{ ft intervals}$

Downstream taper: 50 ft at ~25 ft spacing -> 2 cones (MUTCD Sec 6B.08: 50-100 ft downstream taper); 13 tangent + 2 downstream = 15 cones total

Source: MUTCD 11th Ed. Sec 6K.01

Advance Warning Signs

Speed 45 mph, road_type = 'urban_high' -> Table 6B-1 category: urban_high

A = 350 ft, B = 350 ft, C = 350 ft

Position	Code	Station (ft)	Distance from Taper (ft)
C (furthest)	W20-1	2,560	1,050 upstream
B (middle)	W20-2	2,210	700 upstream
A (nearest)	W21-5aR	1,860	350 upstream

Source: MUTCD 11th Ed. Table 6B-1

Colorado Requirements (CDOT S-630-1)

Check	Result	Citation	Detail
Signs on both sides of divided highway	FAIL	CDOT S-630-1 (July 2026) Sheet 2, General Note 8	Required: True. Signs placed: 6 left, 8 right.
G20-5P Work Zone signs every 2,640 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 4	Zone length: 2,560 ft. Required: 1. Placed: 1.
Speed reduction <= 15 mph per sign installation	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 3	No work-zone speed reduction. Posted speed 45 mph applies throughout the zone.
Flagger station lighting 500W @ 8 ft	PASS	CDOT S-630-1 (July 2026) Sheet 2, General Note 22	Not applicable (no flaggers).

AADT threshold for mobile operations (<= 2,000) — Not applicable (not a mobile operation). (CDOT S-630-1 (July 2026) Sheet 23, Case 38 Note 1)

All Colorado checks pass: False

CDOT Case Reference

Case 11: Shoulder closure on divided highway

This scenario matches CDOT Standard Plan S-630-1, Case 11: shoulder closure on a divided highway.

The generated plan follows the same device layout as the reference case with spacing computed for 45 mph.

Routing: shoulder_no_reduction

Reference:

[https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20\(19-Page%20Set\).pdf](https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20(19-Page%20Set).pdf)

Geometry Validation

Work zone 1000 ft vs required taper 150 ft ($L/3$ (shoulder taper)) + buffer 360 ft.

All geometry checks pass: True

Source: MUTCD 11th Ed. Sec 6B.06 (buffer) and Sec 6B.08 (taper)

Corridor Validation

OpenStreetMap corridor check ran with no warnings.

Site Conditions

Scanned along the corridor against OpenStreetMap at 2026-09-05T06:12:51+00:00 (7075 ms, memoised). The five rule-bearing conditions below are plan facts; a detected condition fired the matching Site Adjustment that follows.

Condition	Result	Evidence
Adjacent at-grade intersection	DETECTED	26 found · nearest 34.1 ft from anchor · unnamed at 39.7038, -105.0816 [outside @ 2732 ft]
Adjacent interchange (highway ramps)	None along the corridor	
Pedestrian sidewalks	DETECTED — dismissed by operator (fenced off)	18 found · nearest 46.6 ft from anchor · unnamed at 39.7112, -105.0812 [lateral 75 ft off centerline]
Bike lane / cycleway	DETECTED	27 found · nearest 71.2 ft from anchor · unnamed at 39.7092, -105.0818 [lateral 78 ft off centerline]
School zone	None along the corridor	
Railroad crossing	Reference — none	
Hospital	Reference — none	
Road curvature	Reference — none	

Operator corrections to the site scan (verify in the field or on imagery before deploying):

Operator dismissed the scan's pedestrian sidewalks: fenced off. The plan is built to the correction — verify it in the field or on imagery before deploying.

Site Adjustments

Site-condition flags layered onto the baseline MUTCD/CDOT layout. Each adjustment is traced to its source rule; the rendered plan, device list, and crew narrative reflect every item below.

Rule	Adjustment
MUTCD §6N.12 p. 848 — Work within the Traveled Way at an Intersection (11th Ed.)	No devices added — the cross-street approach layout is not generated; see the pending-verification disclosure.
MUTCD §9C.101 — bicycle facility work zone signing	Added 2 M4-9a BIKE DETOUR signs at the upstream and downstream ends.

Flagger Control

Not applicable for this scenario.

Source: MUTCD 11th Ed. Chapter 6D (Flagger Control) and Chapter 6E (One-Lane, Two-Way Traffic Control).

Pending Verification

2 reference(s) pending verification. An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. Tracking: <https://github.com/rtmakatura/conestruct/issues/117>

- [intersection_layout_not_generated] An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. (<https://github.com/rtmakatura/conestruct/issues/117>)
- [site_condition_overridden] Operator dismissed the scan's pedestrian sidewalks: fenced off. The plan is built to the correction — verify it in the field or on imagery before deploying. (<https://github.com/rtmakatura/conestruct/issues/224>)